

ActInf Livestream #033.1 ~ "Thinking like a State - Embodied intelligence in

Livestream | 1:50:58

[Music]

[Music]

all right hello everyone welcome to
active lab live stream number 33.1

it's november 23rd 2021.

welcome to the actimf lab

first-time participants and returning
participants and listeners

we are a participatory online lab that
is communicating learning and practicing
applied active inference you can find us

at the links here on this slide

this is a recorded and an archived live
stream

so please provide us with feedback so
that we can improve our work

all backgrounds and perspectives are
welcome here and we'll be following good
video etiquette for live streams

we are in the final two weeks of
november 21

discussing thinking like a state
paper of choice

and we have some tentative paper
selections for number 34 and likely
we'll um take a break and start with 35
in the beginning of january 22

but we're really looking forward to an
exciting 2022 so if you have any ideas
for people to invite for other streams
or if you yourself would like to join
then just let us know

today in active stream number 33.1 we're
going to be learning and discussing this
very cool paper thinking like a state

embodied intelligence in the deep
history of our collective minds
by aval aka serval gwen and khalif
and it's just a discussion video even
though we have the author on the stream
which is awesome

we're pretty much in the dot one going
to
go over anything that we want to we can
return to some keywords and bigger
questions that we raised in the dot zero
we can add notes for next steps for
questions to think about over the coming
week to touch on again in the dot 2 or
possible ways somebody could continue
this avenue of research
and we will be discussing this paper in
next week 33.2

okay we'll uh go around and introduce
our
panel today everyone can give just a
short check in they can say hello and
something that they remembered from the
paper or something that they were
excited about to discuss and then pass
it to someone who hasn't spoken and then
we'll

conclude with the author so

i'll start i'm daniel i'm in
california and

just as we were discussing right before
we started i want to explore this
aspect of states as anticipatory
cognizers not just reactive cognizers
and i'll pass it to

dean

hi i'm dean

i'm in british columbia and

ensconced in a uh cedar box today
out of my cabin
and
i like the paper uh because i want to
see whether or not
there are parallels between what happens
on an individual level what happens on
more organized levels of individuals and
i'll pass it over to blue
good morning or good whatever time of
day it is in your world everyone i'm
blue knight i'm an independent research
consultant in new mexico and i like this
paper for many reasons but
my favorite reason is the scale free
aspect of self-assembly so it's
self-assembly at a much larger scale
than i'm used to i normally study cells
and sub-cellular organizations so i like
how active inference unites
the things that i like to study with
statehood and larger scale organizations
and i will pass it over to first time
participant andy
hello everybody my name is andy i'm in
uh richmond virginia i'm very interested
in looking at the mechanics of how
uh intelligence works through a
collective right uh
are there universal things that we can
that we can apply for
just the movement of information through
um relatively close systems and
and how to modify that
the cognition that goes along with it
now we'll pass it along to stephen
hello
and good morning steven here in toronto

um

yeah i'm interested in this paper

in terms of the scale the social

uh impact of of of looking at states and

how active inference scales are and also

maybe a little bit provocation in terms

of like how much are we looking at

the the

the dynamic states

of organisms or macro organisms how much

we looking at the state of a

dead set of artifacts that helps us

work with our niche

and the scale of working with macro

artifacts in the niche as much as

macro

cognitive agents so i'll be interested

about that um and i'll

um

pass it i believe

to daniel

thank you steven um i think we've all

we non-authors have gone so

several would be awesome just to kind of

recap

hello and

what

brought you to writing this paper as we

kick off the discussion

so hello my name is aval and arseval i'm

training physics epistemology um great

things

and uh what i'm interested with this

paper is to show that we if we accept a

naturalist understanding of recognition

which is a saying that is based on

actual systems actual dynamics

then we can't resist the conclusion that

social systems are organized in a way
that is cognitive
and i found this uh basic constellation
pretty important because
then there are no limits like systems
that are cognitive uh work in the way
they are structured to work and we can't
uh we can change things and we can
understand how change
will affect how things behave
which is pretty good jones because i
think
it's pretty likely that the national
state systems that is dominant today
nation state capitalism system
is uh dying and is uh basically bringing
us with it
so we have to uh explore the state of
alternatives pretty quickly and
efficiently and this
uh needs a reflective understanding on
uh what this database is
great and i think it's um
it's a
a pun or a homonym that will return many
times what are the states because we
talk about partitionings of states a lot
in active inference internal blanket
external nested systems and now we're
talking about the nation states of
um
how are social systems cognitive we use
that state-based partitioning to talk
about other cognitive systems
ones that are designed or evolved
and so
we're applying active inference to study
the cognitive properties of

states um steven and then anyone else
and i'll just keep on adding notes or
questions and people can look at it or
suggest things in live chat go ahead
stephen thank you

yeah i think it's also cool with active
inference there's that kind of
ontological challenge whereby we're
dealing with non-equilibrium
dynamical states in a nonlinear system
which gives rise to pullback attractors
which

have this non-equilibrium steady state
quality

where

however often states are
or say we're thinking quantum theory or
something you know they're like it goes
from

this particular state to that particular
state right so

um there's there's also that
interpretation of what can be done
with states when you start to have
non-linear dynamics at play

thanks blue

so

something that i mentioned uh in the dot
zero video that i'm interested in maybe
exploring a little bit deeper is like
what do states aggregate into so what is
like the grander scale if we're all just
self-assembling into something and and
the quantity of states that are produced
i can't remember who said it maybe it
was a veil that states um like arise as
a result of war
and

faction and division and so as we are
progressing forward in time
there's an increased amount of faction
and division and like it doesn't matter
how

homogeneous a population is they will
find something to divide about and so
like this creates more city-states
within city-states within city-states so
is there some kind of equilibrium like
attractor space for like the optimal
number of states or is it all going to
be like unified or is it all going to be
dissolved at some point

thanks steven

actually that's a good point there blue
there's also something interesting about
democracy they say that for democracy to
work

it almost has to be a split of around
45

to 50

um

[Music]

between different opinions parties
otherwise like if basically one party is
supported by 65 70 of the people
generally speaking

it doesn't work very well because then
there's no there's no dynamic i suppose
to move backwards and forwards um so i
think that's quite interesting

so when we talk about the minimum number
of like states or or

entities to exist for democracy to

function opinions um

it's interesting you know in the system
that mike levin works with back from our

july live stream the computational
boundary of itself he works in planaria
which are regenerating organisms but you
have to have a minimum number of cells
and particularly like a minimum number
of stem cells so i think it's like a
thousand cells at base and like 120
planarian stem cells or you don't get
regeneration and so like what kind of
dynamic is required or how many people
and even for like you talk about hunter
gatherer organizations like
50 to 500 people they think made like an
optimal hunter-gatherer colony like less
than 50 and you're going to be eaten by
coyotes or whatever so
where is that optimal number in
statehood what does that look like
yeah what
aspects of the system are necessary
sufficient incompatible at odds with
certain kinds of
regimes of state and it is really funny
how
we talk about policy selection policy
sounds like it'd be more from
international policy than it would be
from reaching for your cup of water
and then
we talk about regimes of attention
usually people hear regime they're
thinking some historical regime of a
state
whereas regime of attention doesn't
necessarily carry that connotation
so linguistically there's a lot of
contact points and that's why i think it
was a awesome contribution by cervel was

to just draw those out ground it in the
physics of critical systems and of
cognitive systems and
that's what is contributing to hopefully
some of the ways we'll explore it's
going to be
used going forward so dean and then
stephen
yeah this this might sound like pushback
or challenge or whatever but i it's just
i have a personal issue with the entire
idea of scale free
like that there's some sort of
invariance across scales and i i think
that like when we see for example a
lower case a
morph into an uppercase a
even though they are of the same
categorical type
and fit in the same place
in the alphabet there is a there is a
change that goes on and so there is a
scale
variance aspect to this that i think
so again this is my my sense of it as
opposed to sort of
the agreed sense that there's a
scale-free
process in play
the other thing i would
i would wonder a little bit about is
when we
take a constructivist
view
a city as built
i'm not sure what happens to the
bi-directionality
that

active inference

affords so

again i i'm not saying that

we can't find

qualities

in both but i'm not sure that a city

as a reflection of something

constructivist is necessarily how living

things reflect

and predict but again i'm open to having

my mind changed

thanks dean stephen and then all have a

comment on the constructivism

i was just also i think it's a good

point we blew about the uh mike levin's

work and and the

question of um yeah how things are

scaling up and of course the big thing

that mike levin did mention that you

know they they've got these quite a lot

of ideas around how things grow multiply

and relate but no idea how they stop

and that's kind of also interesting in

terms of you know when we think about

scale as like

has the organism reached its point

because the top scale the maximum scale

has kind of been reached or is there

some sort of

is it that there's it's the dynamics at

a mid scale which is like an action

scale

so it's not it's not that it's ever

arrived at this thing it's just arrived

at this

meso action policy landscape

which is

um available uh and

you know is that availability then
understood from what's inside or is it
something that's checked to something
external in the environment so
that that i think is an interesting
question

thank you

um capitalizations yeah you have to the
state capital capital punishment
there's so many um other funny word
plays but to the constructivist comment
so

um

one sort of simple constructivism would
be like i mean there weren't bricks
there and now there are so the city was
constructed and now it's the scaffold
for us to interact in so you ask dean
where's that bi-directionality

so i think there's a few

ways to talk about the bi-directionality

so first is niche construction

can be a symmetric relationship we talk
about the agent

modifying the niche and the niche is
modifying the agent and so it's like you
have the blanket as the interface and
then there are

action and inference on both sides of
the blanket being used as an as an
interface and so it's more like a
co-constructionism

but if you know lego set a is building b
while b is building a that starts
looking more like interactionism

because

it's not just instructions put the brick
here

it's a specification of a protocol and
an interface for interaction and that's
what opens the door to improvisation
into other kinds of dynamics that aren't
just simply like assembly of the 3d
printer and then the other point and
then dean is it reminds me of dennis
1995

book darwin's dangerous idea
where he talks about basically two ways
that you can talk about the appearance
of design
in natural systems which are cranes and
sky hooks and so if you look at a
skyscraper you think how could that have
appeared one approach is they built a
crane to put that last top piece on and
if that crane was too big to construct
then a smaller crane built that crane so
you could have a cranes all the way down
like small things were built and
scaffolds were in place that were
removed later to result in this
skyscraper

or you can just cut the knot and just
say well maybe there was a sky hook and
they just latched it to the sky and then
built down so that's like top-down
constructivism
appeals to skyhook technology whereas
bottom-up
emergent process-based co-constructivism
is more of a crane based approach
so i think that's a very good point how
even with constructivism we can still
retain a lot of the important aspects of
constructivism like physical
construction but we don't need to fall

into a
uni-directional um or a top-down trap
so dean what do you think about that and
then stephen or anyone else
i think i think the important point
daniel is what you just mentioned which
is the trap of
prematurely collapsing to the top-down
privilege um
for example uh
if you go to if you go to just outside
of denver
you end up in a place called red rocks
and then the question becomes did
did uh did somebody enact red rocks
because they put a stage at the base of
an amphitheater or did red rocks enact
everybody who
was attracted to that place and so all
i'm basically suggesting is is
is
when we look at how things are
constructed we don't want to lose sight
of the fact that sometimes
things construct us as well sort of the
sky hooks aspect of it
thank you dean stephen
yeah and also i mean cranes and sky
hooks
it gives that idea of how it links in
they're also dead items right so they're
both tools or they're both objects
um so
and we can we can bring in
sky hooks which might be
informational instructions or cranes
which
are

you know places to carry the bricks that
are going to build things but i think
that when we talk about that markov
blanket and what goes both ways is
yes i think when we're dealing
particularly with dead things
like bricks and that we construct an
environment now it's a bit more
complicated when we impact an
environment that also grows and changes
and is alive

however when something comes back the
other way from the environment is it
constructing us or is it
scaring us or is it giving us lots of
little

punches in the gut in which we heal
so i'm wondering whether the other
direction

it can't construct us

but what don't kill us makes us stronger
type of thing

thanks steven

um andy please

and then blip

dean i think in the context of

cognition in the state

the specific um the element that's

missing right is what's the relationship

between the nation and the state right

so the nation would be the bottom-up

information or the bottom-up cognition

and the state could be theorized now as

the top down

and then broadly speaking right the

nation creates the state

the state the existence of the state

changes the circumstance the changes in

circumstance forces the nation the
nation the people
to rethink how they're operating in the
world around them because their
circumstances have been modified and now
you have this constant bi-directional
relationship between bottom up and top
down in the context of state versus
nation and modifying circumstance over
time

thank you

blue

dean go ahead oh yeah dean

yeah andy uh

i like what you're saying except

it's already collapsed to an a to b

relationship

and i think what active inference allows

for is to keep things up on it on a

generalized abstracted level it doesn't

have to be here's the nation a and

here's the people b

that's the difference

essentially is if you want it to be an a

and b thing

go ahead that's that's

good for argument's sake but in fact

what active inference does is it keeps

one side of this open

and not necessarily specific

it keeps it

in order to be predictable it asks the

counterfactual what if

as opposed to having that

pre-presented as the relationship

and it's new

it's a new type of relationship that's

arising and i think that's right

explored in the paper in the deep
history section there's really the
specifics there's the minute particulars
of the specific states that we have
and typified by westphalia and a bunch
of other historical events that the
specifics matter we're in a complex
system the trajectory really does matter
blue

so i totally forgot what i was gonna say
um

but i would like to maybe challenge andy
like what is the the difference between
the nation and the state like is the
nation like the people that make up the
state and then the state is like the
collective of the people like is it like
a a part whole

concept or are they two separate uh
larger entities and people make up both
um so
words

depending on which definitions you go
with right it gets a little bit squishy
in the context that i use them
state is a construct

the political organization of a people
is

is downstream of uh the shared identity
of the people

of a people so the nation would be
broadly the largest sense of shared self
distributed among a relatively close
system in this regard and the state
being the construct by which these
people organize their affairs inside of
their nation

that's a relatively new idea right the

idea of a nation and the nation state is relatively new as far as like an overt construction um beforehand in in the age of absolute monarchy there was no separation between the state and the people right it was all embodied in the in the person of the sovereign um so different different historical contexts different periods of time different cultural context now you might get uh different definitions but in the way in which i was using it the state would be a policy requirement of the culture in which it was born right it's why american democracy functions significantly better than south sudanese democracy even though they have literally the exact same institutions so daniel maybe you want to flip to slide um 33 i think like where we we like kind of show in the paper all of the like things that go into like the production of a state and so i like i'm curious as to like where nation fits into this um is it like the in the collective identity or is it like um a relationship between like nation and collective identity or just what your thoughts are on that

yeah absolutely collective identity to
me would be the definition of a nation
shared sense of self
everything else are things that you may
or may not need a state to accomplish
that may or may not be more efficiently
conducted in the auspices of the state
right um control the monetary flow you
probably need something as sophisticated
as a state but to transfer resources
back and forth you don't
right you can you can grow grain and you
can you can trade it for for me
uh go ahead
well just the the um collective identity
i'm just thinking about in organism
and the liver and you know the bone cell
it's like
collective identity doesn't mean
identical phenotype and so it makes me
think of a cultural memory where each
person maybe there's a few touchstones
that 80 percent of people know about or
increasingly though there's less and
less shared information
amidst
the informational changes that are
happening so what does that mean
when everyone's only going to have a
little partial facet of the collective
memory
on board
but maybe they can enact broader
processes that are truly collective so
how do we respect
differences
and
shared aspects across multiple levels

so um blue and dean and then serval want
to hear what you're
thinking about this
so just like daniel how you were talking
about like the bone and the liver cells
like making up an organism um you know i
think that that speaks here to what's um
the specialized roles
and like the density of settlements
enables that like functional
specialization and
like you can have a blacksmith and a
shoemaker and a farmer and everybody
takes on their own little role and then
goods can be redistributed so it just
like reminded me of that and like that
functional
specialization um and the importance of
it
thank you blue dean and then cerval or
pass
yeah when i was watching the 33-0 and i
saw this slide thank you blue for asking
for it to be um
brought brought back again here this is
again an uh a specific to a specific
because it's historical and it's
reflective
and
the fact that we can look at things and
say this is what happened we can look
back on them i'm not sure we want to
necessarily conflate that with
active inference which is about the
recursion
so
this is what happened we can see
how things moved across a timeline from

point a to point b a city state evolved

but is that recursion

which is what active entrance is about

which is variation upon variation

i'm not again i ask

is that

i know how we can i know how we can

diagram it and represent it but is that

representative of what actually occurs

in recursion when certain thresholds

appear and then just as quickly

disappear

interesting so

serval if you'd like to add anything

and then stephen

so there are a lot of threads that were

opened what are you asking my opinion

about

um

[Music]

i think part would be just uh on

nation and state similarities

differences um what is and what if

and then more on this sort of

integration

slide

where does active inference come into

this tying of the knot

how does acting in france

uh about nation i don't think it does

nation is simply a collective identity

that is hated to a political entity it's

a

community of

peers that happen to have some stakes

and power over the behavior of the

institution

material states

and the fact is that nations are not universal you have a lot of states that are not nation states that are not suited to a community of peers for example

the roman empire it was a empire which means that you have people who get the power that form authentic group that is the roman and you have all the rest of the people that exist in there that also are ruled by the um by the empire that is dominated by the romans so you do not have a community of interests that illustrated to a state you have a netting group that dominates other ethnic groups through a state the nation are not um they are not invariant aspects of what the state is however they are um a collective identity or at least ways to build boundaries between a group and ad group there seem to be an environment of efficient human collective action according to astrum's principles so there may be more tendency to build nations to the nation states uh when you have more uh importance uh between group selection and it is said that i don't know of any uh computational quantity work that uh but he said that modern europe is pretty

extreme in terms of um
of
between group conflicts of war
because
the diamond agreement is that the
geography basically enables trade at
great scales but prevents
the constitution of an empire
in
europe
so
historically there were a lot of states
in europe as compared as say uh
in china where you have a big state that
is
i don't know
in terms of surfaces maybe uh two or
three times like western europe is
and more integrated than when you can
have in south america because of the
mountains
um so you have a pretty ideal regime to
get a lot of intergroup conflicts and to
get a selection regime that favors
nations uh organizations that somehow
collate with states and that's what
happened to be instituted through uh i
don't know westphalia and french
revolution i don't know the story of
that there are a lot of parkway pathways
in europe from the states but in europe
you got nation states originated you've
got pretty aggressive kinds of
imperialism that originated
and both can be traced i think uh to an
intense regime of uh between group
selection
and of war

and
there are results that tend to say that
war
explains in the sense predicts
more political complexity which is
uh
a debatable construct that basically
maps
this kind of integration of society
so
the uh
history of europe
of modern europe specifically um and
meteorology everything past the roman
empire
would have uh selected for both
nations
and
aggressive kinds of imperialism and this
is the kind of organization we get to
live in now
thank you
serval just like the ecological
regularities through evolutionary time
scaffold the evolution of an organism's
cognition
this is a transposition of that argument
to the biogeographical
and contextual and ecological factors
that give rise to and scaffold the
development of states as things as
cognitive entities there are some
parallels there so it's an awesome point
thank you blue and then stephen
there is a
pretty important things trying to say it
that is a state does not actually
correspond to a nation maybe states have

original nations maybe they are
assigned nations now but a state need
another entire nation you know you
cannot states
like cultural population united states
uh
well that's kind of
the usual deal actually
so
yeah election is a collective identity
of
in a community of interest states uh
control systems
uh built from administration and it can
be and often is captured by an elites
that is um technically
social like socially in terms of
cultural norms lobes pretty distinct
from the people controls
chicken in the egg
egg as the secreted
state
chicken as the life form but then
there's a co-evolution so that paradox
stands there too blue
steven and andy
i'm gonna defer and come back around to
my idea so stephen and then andy
yeah i think
the the whole
gale question is very interesting here
um and um
where it sort of so often i mean in the
european union you've got the european
union it's kind of a reunion of
pluralistic partners is the idea and
you've got this tension where one side
is basically saying no you're not you're

a state

and we want to be a nation

an independent nation there's this

tension right between oh no we're a

union we're a union

and when actions are working that works

the union right and uh so i think that

there's a scale where people try and

work at and and the rate at which people

work at

what's the dynamics of people

in the same way that when you have um

politics seems to have a very slow

trajectory you know things shift you

have the state you have your end

identity of your party however nearing

an election

a week this praise is a week is a long

time in politics and it is near an

election

the dynamics start to become much faster

things can change much quicker

so this scale question again comes in is

what is

what is a unit of identity how does it

manifest at different rates

um

and just one other thing in terms of

identity is in

participatory

community-based action research

okay they talk about community is

actually a unit of identity it's almost

like there's so many different types of

fluidity at the community level and

overlaps between communities that it

becomes a unit of identity to actually

work with in measuring stuff

we tend not to do that like at the
higher level the nation it's just the
flag is a unit of identity
it's a very which hasn't maybe changed
in 100 years so it's an interesting
um as well as the you know what how much
can things change and how much are the
ability things to change caught up
in the artifacts that embody them
the artifacts that constitute them in
some case

so andy

any comments or to blue

yes thank you um

so

i was going to well let's let's go deal
with what you just said right community
smallest action uh element of action in
this context

uh identity is the smallest
element of action

however it is guided or bounded right by
its ability to affect change

so if you if you deeply identify with
one other person in this world the two
of you uh are the small that you have
you

identity becomes basically shorthand for
interest we two people by calling
ourselves a group have a certain
worldview and a certain
desire to see our interests achieved we
have the capacity of two people to
accomplish that

so what is our ability to affect change
at that scale relatively small once
you're talking about anything larger
than interpersonal relationships right

but it doesn't change the fact that your identity is shorthand for your interests and that we all at whatever level or scale we're operating at are pursuing our interest to the best of our ability bounded by relative power right um but this idea of of nation versus statehood survival you're absolutely right the state is a thing independent of a nation sometimes they correspond and typically we say that that is um that that's a good fit because otherwise it's an empire right otherwise it is somebody else's ideas of what constitutes uh right or wrong in terms of projected values or institutionalized values that are now imposed on you because the state is not representative of your cultural world view or your shared identity it is something provided to you by somebody else's in the form of empire but the question that i had right going through the slides there was this idea that the state as a top-down hierarchical control system what is then nation or what is an identity how does that function the sharing of information and cognition relative to the state thanks for the uh text-driven question it was a core argument of the article just to quote that states can be and should be understood as hierarchical control systems so

um avail if you want to give a direct
response otherwise dean in blue
so there is a direct response
uh which is that
uh to talk of hierarchical control
systems specifically to talk after done
control this is
i think i've
reframed the same uh point the same
thing in saying those both uh words
and of course there is no cognitive
systems that is purity of thought but
there are they do not work they live in
labs and they are not useful
uh so
basically uh the organic
organization of whoever happens to have
a say in what the states is
is the
at least the ascending dynamics in the
collective when they are what affects
uh in bears in terms what uh change the
priors and the policies etcetera
so
in the case where you a defined nation a
defined entry group
that happens to control the states
which i'd say is exceptional from what
i've read
you can't say that there is a meaningful
nation states fit and
the nations basically the brain and body
of the state in the sense that
well the ascending dynamics that
influence what the control systems does
in most cases you will
have i'd say a pretty much stronger
decoupling because basically only elites

would get to say

in

what the state does and in cities where

in mercedes the elite is not to define

an ethnic group

uh

in that case you will not have a clear

fit between nation and states although

um

of course the committee of power the

defender states if you conceive it as a

nation

will uh define what the state's prior

are

well pacified to define because of

course there are the tools that register

thank you so awesome lot of hands so

dean then blue steven

um so i want to just

circle back to what andy just said andy

when you if if you go and look at

this live stream

three hours from now even when you see a

perfect reflection we'll assume that

this youtube video comes out exactly as

as it intended

i think what you just said

proves that we're not talking about

scale free

talking about scale friendly

because what you will see

is not what you just did

and so i think that that has a huge

bearing

on this idea of what we're talking about

here we can shape-shift into what

stephen

cilit has pointed out here is that that

fact that we can

fill

between these two levels of scale

doesn't mean that it's it's it's one

size fits all

it quite literally means is that we can

fill we can change we can modify

and i think that's recursion in its

essence as opposed to

taking a reflection

and knowing

that the next time we look at that it's

going to be different because there's

some elapsed time between the moment we

set it

and then the moment we heard ourselves

say it

dean i think it's an awesome point with

scale free and scale friendly skill-free

people usually mean to say that the

mathematics or the formalism doesn't

have an a priori system of interest or

scale like this is a centimeter theory

or this is a kilometer theory so it's

scale free because it doesn't have that

a priori reference scale but then scale

friendly it's almost like how about you

tell me what scales you've identified

and let's make them friends

through interaction and iterated

modeling and so we're friendly to scale

and scaling

but that doesn't necessarily

mean it's free although from the other

side it's also fair to say that it's a

scale-free framework in the way that a

bayesian computational framework or even

a linear model is scale-free it doesn't

a priori say that kilometer is the
default unit for linear regression
either
blue and then stephen
so i think what i'm going to say is
going to touch on the comments that were
made by both andy and cerval um first of
all in terms of this relationship um
like the mutual interest
being the fundamental unit of identity
and like collective identity this really
goes back to
expansion of your cognitive boundary
beyond yourself and like when we were
talking about the computational boundary
of itself again i'm gonna like always
refer to that live stream but um there's
an informational sharing
when there's an identity sharing so when
you expand your cognitive boundary to
include more than just yourself like if
you have like a really deep connection a
really deep relationship with one person
you have like a a great degree of mutual
information between you and that one
person and then so say you have like a
family unit or like a network of close
friends there's a lot a large degree of
informational sharing
and so in this
sense of building a nation
um like the fundamental unit of identity
then maybe is is actually informational
so it goes back to like bayesian
principles where
there's this informational sharing that
that then allows you to update your
generative model um so so just to kind

of speak to that and then also maybe um
to
uh
just like maybe help andy make the link
like between
um downward causation imposed by like a
government or like a hierarchical
control
institution and like the the emergence
that happens like when people come
together and share information in a
collective there's a super cool paper by
david krakauer called the information
theory of individuality and it talks
about like in order to become an
individual and there are many different
kinds of individuals but in order to
become an individual there has to be
like a bi-directional information flow
that's both emergent starting with like
the fundamental units like the people
sharing information there's an emergent
and then a top down so bottom up and top
down and like when you have this
bi-directional information flow that's
like what allows you to
form a boundary i guess around your
collective
something like that
thank you blue
steven and then anyone else
yeah i um so picking up what dean said
there and um
and actually blue and everyone else is
the formulism is is scale free at the
free energy principle and then you get
into applied active so then you get into
active inference

which has a sort of scale-free quality
and then you get into adaptive active
inference
and then you get into
you know
inaction
of adaptive active inference and then
you get organisms like us that want to
apply active inference and i think that
i think it does like
really start to go scale friendly
once you start to go up
um and and one thing that i find
interesting i've been this actually is
nice it ties into this infusion space
work that i do but i look at the face
and the body so
at there's a particular scale
at which my mind is embodied
and you can see it
and the bit that you see is the bit that
we show on this screen
you're not looking at my foot or my
shoulder
or my hands
you're looking at the face
so the face
is the embodiment
of my
mind as an identity
okay this is my projective identity and
it becomes very useful
so it's it's scale friendly at that
scale but you know until
the other scales of my body aren't you
know my liver doesn't do a great job of
my identity okay and then we got okay
what scale do i have to get to i'm

talking around if i'm just talking
one-on-one i don't sit here waving a
canadian flag or whatever right now if
i'm a crowd of a thousand
now maybe we need to have a big flag on
a stick right and that becomes my face
right that becomes my projection
but that's a scale friendly because now
i've again it's kind of gone in and
recapitulated the environment
so that's kind of interesting
um thanks
thank you stephen cervald
i
don't really get the debates about uh
scale fiscal friendly thing
uh because what a scale-free theory is
is a theory that is
in principle a formalism that's in
principle it's not a really different
scale so for example my opposition to um
the weak and strong forces that only
have a
measurable effect at very small special
scales uh gravitation and
electromagnetism they can influence
things but that are pretty much so far
well not release money some because you
have the um
um
cons mutual counseling of plus and minus
charges but in principle it could
affect things
at distance because of uh the error
square
the repentancy just kidding
and uh the fact that you have a theory
that is capri does

not entail that every system
where you express the theory is calculus
for example you have the
gravitational theory the skeleton but if
you study the solar systems it has a
scale which is about the distance from
the sun and the uh
thing apart if you study
electromagnetic motor it has a scale
which is uh well the size of the matter
and if you study a biological organism
with a scale pre-formalism it has a
scale which is whatever scale at which
the organism happens to exist

so i don't

get

what the debate is about

[Music]

dean

how about

give a thought on that

yeah i got a thought on that so i think
we can see what a reflection is we can
see what a triangle is we can see a very
small triangle and we can see
a very large one those are reflections
but that is not recursion
and as soon as you get into
variability upon variability you're
going to have to come up with something
more than a reflection
and i think that's why you have to come
up with something that more than
scale free

i mean we can see scale free and
nobody's arguing against that but what
was i guess what i'm pointing out is
that if we limit ourselves to that we're

not going to see the entire picture
all kind of
intermediate i think they're both very
valuable views scale free is certainly
more aligned with the way that people do
talk about formal models whereas scale
friendly is a little more of a dean-ism
which is always fun
i see skill friendly as
emphasizing the active role
of the modeler in connecting the levels
so it's not enough to just say like i
have a stretchy ruler that could apply
to different scales it's not an a priori
scaled ruler
and then making it more of this process
of approaching
the mechanisms and the processes that
are interacting
amongst scales rather than just thinking
well we'll put the linear regression on
a little thing we'll put the linear
regression on the big thing and then
we'll have a scale-free applied theory
this is more like let's take that system
as a indivisible unit
of function and then like interact with
it in a little bit more of an active way
and it can still be scale free for those
who like that
way of talking about it
um but for those who want to get to know
it maybe scale friendly is approachable
as well
stephen
yeah i think
there's also
active inference and adaptive

active inference and inactive
influence so i think like daniel made a
good point there is
uh
keeping the active influence flowed
through and helping that
is really useful right so that we don't
lose the formalism side because that's
the danger right we clip off the
but but there there is something for
instance
at what point is adaptive active
inference possible like is that the
point when you need to have the brain
within a reason or there needs to be
there needs to be some
generative process that's regulating
active inference for it to then adapt at
different temporal scales
um now that that raises some other
questions about exactly where but
there's
that you know even the idea of what
adaptive active inference and active
active inference or adaptive and active
potentially is
starts to
then tie in well you know and if i can
only do certain adaptive active
inference with the brain then has to be
the scale of an organism that has a
brain
and below that scale it can't so there's
something
that becomes transitional or
contextual
dean on this point and then to serve all
for a combined response

yeah um

so again i think

knowing that a triangle is a triangle

regardless of the volume

of its footprint is important i think

when we include updating

when we include time

then we should include

scale friendly so i'm not discounting

scale free

but i'm saying that the updating process

whether you want to call that adaptation

or in action the change aspect of it

doesn't get resolved simply because a

triangle is a triangle

if that makes sense

bucky fuller said angle is is

metaphysical and general it doesn't

refer to any specific substrate to say

that something is a 60 degree angle and

frequency is always bound up in space

and time and the specifics

so any realized system is going to be

both

serval

so um

stephen i think talk about an active

acting in france rtx in france is uh

something that keshav introduced in 2018

i think something like that

uh tries to reproduce inactive notions

of cognition from active influence what

it says is that you have

agency when you have a hierarchical

modulation of

active influence system you have higher

level of action fronts that modulate

lower levels of active influence

the question it rise is uh basically a
question that is
not solved and i don't even know what's
happening on it
in the ontology of
science and naturalism which is
do you have different level of existence
or do we have only scales
and even at this level when we admit
that we have things within things within
things within things and that's what the
world is
it's complicated
the thing is that
uh to circle back to a question that has
been raised initially about how what in
france relates to other values of
cognition
um
the while keyshot
tries to formalize inactive concepts of
cognition is entirely wrong
because in activists they do not say
that
organisms are heretically modulated
acting from theorists say that the brain
is killed modulated and they apparently
projected its own activists but we say
that
a system that is cognitive must be
operationally closed which is it has
set of
it has a structure
uh or an organization i don't know which
because of their consumption uh that
basically reproduce its own existence
without any clear support without
autonomously from its environments

uh that's not how they say but whatever
and you have a specific formulation
that's computational it was formulated
by um
people from the basque center of life
sciences i think the name is
uh which is uh constrained closure so we
have constraints that are slow processes
that shape how uh faster processes
behave
and wealth a lot of them and
collectively this lot of constraints
shape faster processes so as to maintain
themselves that is um
constraint closure and that is
what's the latest version of um
on the elective um
not agency but autonomy nation is
and this entails not hierarchical
modulation but uh meshed nets of
processes that make something else exist
at in between scales
and the notion of what is the antics
that choose
to give this is pretty complicated i
don't know i a lot of people that think
that level is just an epistemic
construct that is built by scientists
and this is pretty consistent with a
radical approach with a mentioned
approach
this
everything becomes ambiguous the meaning
of every single terms
become vigorous so
i have um
no idea what will be a correct way to
refer to this

uh scale free says what it says that it is not interested in specific scales of behavior and uh it also says implicitly when we are in the context back in france that it is not interested in a specific antique level what applies for ourselves applies for a mutual organism which applies for societies and which in the case of vanilla in france also applies for dust vertex flames and whatever so i'm not sure we're still and i'm still not sure what the national scale friendly is supposed to do and uh how it relates to uh the question of whether states are properly cognitive because we don't need a proper version of population of scales and proper anti-distinction axing distinction between what is uh recursive and what is only a scaling variance we only need to state what being cognitive is and to check whether states are and of course uh what this means for the broader uh episode constructs with builds around this thank you sir blue then steven so i have to like go back to um the active inference live stream number 30 we talked about how to count biological minds and really this like raises so when you're talking about the distinction between adaptive

active inference and like mere active inference

um and in this this theory about how to count biological minds is everything that's biological is do all things then have a mind like even down to like the you know single cell organisms and in this in this paper like there are a lot of people who like have this like mind life continuity hypothesis and a lot of people really um ascribe to that so in this adaptive act of inference versus mere active inference i couldn't find the line like i really pressed on matt sims about this um and and he responded beautifully like he also couldn't find the line so at what scale or what level does adaptive active inference transition to mirror active inference

and what i'm fundamentally getting at is like okay so are you doing mirror active inference if you're a single cell and you just move forward and back like towards a repulsion or an attractor like is this mirror active inference or is this adaptive also

and so like adaptation to the niche like to me i think that happens at all levels

but there was this distinction made in that paper that like adaptive active inference happens when the collective forms some like emergent behavior that makes it more fitted to the niche environment and so like where is the state then a biological mind does the state make

um a set of organisms like more fitted
to its niche
through this like collective adaptive
behavior
or or is it maybe maladaptive
because overall like as a species the
more divides we have like the more war
we'll have i don't know
thank you stephen
yeah thanks for this meshed nets
idea i think that's that's definitely
true um and uh
let's kind of fix in with the kind of
meshed architecture
um i i wonder if
like um blue was mentioned about this
different scales of active imprints um
identity is quite important here um
and
the nation state um
is a kind of
is is maybe identity driven and i kind
of wonder if we just scale that down i
was thinking scale down i was talking
about the face there earlier but for
instance
the baby in the womb okay
is the the release during
childbirth of the hormones which
triggers the woman to break her waters
basically like a broadcast
from the identity
of the womb so okay body i'm here now
i'm a child bearing womb
i'm declaring myself
in the same way that maybe the pituitary
gland but it does it all the time
declares itself

certain times to crash out into you know
create insulin so could we see that then
this meshed architecture comes at a
scale where it's possible once the you
know and then it's like
and in the same way then
uh the nation state maybe as we go up in
scale
the expression of
uh signifiers
to broadcast out becomes greater and
greater
you know becomes a bigger and bigger
part of it and maybe
the biological
processes become
less a part of it
well definitely it's the question of
realism and instrumentalism whether it
quote is a broadcast or whether we'll
just choose to model it as
so one of those is definitely safe
territory and the other one is
a little bit harder to make claims about
what the territory is versus the map
that we choose to draw around the
territory
um but very interesting
points dean
okay i'm one i just want to make sure
that
i'll get feedback from people on this so
to me adaptation
okay adaptation around what is
which can be scale-free
or if we're talking about active
inference do we also include the counter
factual what if

and i think if we include the what if
then
i'm not sure
that adaptation
looks the same
it's not reflective so i'm not pushing
back on serval as much as i'm asking
are we are we over fitting are we saying
that recursion doesn't matter and are we
saying
that scale friendly doesn't matter
because we only want to concentrate on
what is
that's my question
dean i'll give it a thought then stephen
and then anyone else
active inference agents
that are capable of doing adaptive
active inference as a title so just a
recently proposed word as serval brought
up
their internal generative model includes
a model of
perception and action
as well as of cognition so that's
metacognition
the what is of metacognition at one
level can be a what if at another level
what is is
the agent can actually be imagining a
counter factual
with respect to perception and action so
cognition
is that
friendly scale
it helps us bridge
the what if and the what is
because um there's als there's so many

other angles one other path to walk
through that would be like um in anil
seth's book being you
there's a great discussion about how
counterfactuals actually are like the
basis of cognition like
how do we know that we're looking at a
real mouse
like on my computer table versus like a
picture of one at the same resolution
because actually there is an implicit
counterfactual i could move at a
different angle
and see something different the physical
object but i wouldn't expect the shadows
to move on this jpeg
and so actually
the difference in our perception between
two things that visually can look
identical relates to deep situational
awareness that actually
blur or at least
reimagine the relationship between what
is and what if
stephen and then blue
thanks actually this paper's serving as
quite a useful bridging
uh provocation um because as you just
mentioned there daniel with them
okay so what if you know way of what if
and counterfactuals
so there's there's the cognition which
is a particular scale
of perspectival
or even implicit what if the counter
factual but
what if this and that happened in the
future well the other question is

what am i becoming

and actually if we want to look at scale

3

if

active inference is based on free energy

principles and free energy principle was

pretty much the only form of nonlinear

dynamical system out there but now we've

got some of the statistics stochastic

chaos work um also

um like if you go right back to

non-linear dynamical systems and

a lot of discussion around non-linear

dynamical systems people are talking

about thinking about that as a process

of becoming

so

how because it gives you a way what

dimensionality can use well the the

process of becoming starts to

tie into that

um and

cognition as we see it doesn't

maybe the process of being

is more approached of becoming but the

process of what i broadcast if using

that term very

brutally but what my identity conveys

is a much more

steady much more of a

an artifactual state if that makes sense

especially if i take a photo of it and

put it on my profile on facebook

thank you um here's an article on being

and becoming and that distinction as

well as i'm sure a panoply of german

words has been discussed for a long time

about the relationship between being and

becoming and

just to give one thought before blue

that generative model of the active

agent includes what kind of agent am i

what are my expectations preferences

affordances etc so

we get in the deep active inference

framework built in

what if and then what and then who would

i be and how would others see me and

what would they think if i were to do

this and then this were to change that's

all part of the state space of what's

possible

these are like the grammar in terms of

the very austere outline of perception

cognition action and external dynamics

that allow composition

and new design motifs for those kinds of

questions to be asked

blue

so if you wouldn't mind flipping back to

the twisty slide i think it's 33.

um and when you're thinking about like

are we talking about

what is or are we also including the

what ifs in the model like

when we talk about what it's like to

build a state

so so this is like how states have been

built before

but but how can we like take this and

like as as daniel was like referring to

at the beginning of the talk like how

can we take

the what has been

and incorporate like what if via active

inference to predict

like how to optimally optimally form
states in the future so is active
inference incorporating the counter
factuals are we going into what-ifs or
or like what about what's next like not
just what if but but what's next to
predict like the the cognition
essentially of a state
and how it's it's going to think going
forward
great
question and what's next we can have our
expectations is we can have our
preferences
they can interact in a different way so
dean then andy
yeah blue i think that ties in with the
point twos that we tend to do
all i would all i would
all i would add to what's been said
is that if if we're looking at as
[Music]
as is we're looking at this reflection
and and can we ask off of that of what
are we becoming of becoming which is a
function sometimes of scale many times
of variation
so i know i keep beating this horse
because i want to include the variation
and the fact that time changes
our picture
that's that's all i want included in
this
and i don't think that that
as
is a perfect copy as we update
so i maybe i'm maybe i'm explaining this
wrong but i think as change occurs

that perfect reflection that perfect
copy doesn't move forward without
variation
concentration sometimes sometimes we
just
toss it out the window
and that's the what if sorry thank you
dean andy and then i'll give a thought
on that as change occurs
so uh in my travels i've run across this
idea that somebody tried to
quantify functions of a state
that are universal
right what are the things like how do
how do you maintain health care not
necessarily which way is better but
distribution of resources
education health healthcare
things that we associate with the state
maybe not every one of them every state
exists or every nation exists or let's
stick with state maybe not every state
does every one of them certainly not all
of them do it to the same degree of
competency and they are all bound within
the expectations of the people who
um comprise that
that polity or that that entity
but to get after your question blue
right if we if we want to get to
what if how is this different how does
states form differently what is the
optimal thing
then i think you've got to get down to
like um a core set of variables for
which states are are
supposed to
uh provide

solutions to

or address

um

and then from there we can see well what

if they did it this way what if if they

prioritize this and not that what if

they prioritize this function over that

function and why would they right what

are they responding to

i don't think that they're responding to

um

necessarily uniquely physical or

material requirements i think there's

some interplay between

um

intellectual or

values-based requirements as well as

material requirements all mixed up in

time and space

given

which parts of history we choose to

preference

in a particular moment

thank you

so one note on the as change occurs

so

here's a very influential

book with a a broad

follow-up lineage on the geometry of

biological time by winfrey and it brings

us back to the discussion of kairos and

how change occurring

in chronos

we become like pro trustees we have to

stretch and pull and change

the sequence of action to fit onto

the scale as defined whereas

thinking about the space of action

the geometry of biological time is
defined more by action
like embryological events rather than
how many seconds have ticked on some
atomic clock in some bunker
so
that um as change occurs
can be through time but it may not be um
the kind of time that gets ticked out on
a clock
and that's just a very different way to
talk about actions and inference because
it's dynamic but it's not just like okay
the clock's running and things are
changing it's like there's actually a
timeliness to action and so we're
talking about the dynamics of the
context and the timeliness
and then maybe there's ways to map
kronos and kairos
or maybe one time we play more into one
or the other
so dean and then anyone
else yeah just real quick thank you andy
because i think what you did is you
applied a partition to preferences and
priority and you made sure that we
didn't say that they were one and the
same
and so in real time we we see we see an
evolution
throughout the discourse so i think
that's
that's a real example of what kind of
happens as we update
so one other random
thought just from earlier and then of
course anyone just bring up a thought or

anyone in the live chat so we were
talking earlier about like capital with
the ol and capital with a l
okay so this was pretty interesting
capitol with an ol
comes from capitolium the name of the
temple of jupiter at rome on the
capitoline hill
so
dating in english from the 14th century
so that is referring to like this
building that's like very much like city
on a hill
and um
city on a hill is used as a multi-scale
allusion
to the united states at the very least
um like capitol hill but of course
broader notions of what
the state stands for and then there's
capital with a l
capital on the other hand has a far
wider range of meanings in application
the word comes from latin as well from
kaput meaning head
and then it has a lot of um polycymous
uses
like capital punishment which we talked
of being the seat of government
like sacramento is the capital
of
california but then like venture capital
capitalization
so that's just interesting um
i hope it's not simple word play when
there's a homonym but it's very
interesting that they both draw back to
rome which was alluded to earlier when

thinking about the seeds of what today's
states are on the world stage so andy
then stephen
yeah sir
would institutions of the state right
like um ministry of foreign affairs can
be considered an example of embodied
intelligence
in the context of cognition for
the state
yes
something that blue said earlier
is that
people proposed a collective agency to
be based on the ability of a certain
kind of
regime of her behavior
to
influence individual fitness
what i would say is that
this is wrong
the biological organization it is by
definition organization it
is constraint that exists upon the
collective behavioral system
if you look at say its steel beam it is
a structure it is something that affects
how the steel beam
interacts with the world
and this is something that is entirely
invisible from the perspective of a
single particular steel which is the
natural lower level
lower scale
perspective you can articulate
so the view that
organization is entailed by
things that affect the niche of their

constituents

is flawed what organization is

is a constraint at a specific scale and
specific

behavior on a specific dynamic

and

the relation it does

with the body

the body

which

certain uh certain schedules of

cognition is about

is uh the sets of those constraints uh

that collectively enacts uh behavior

that we say is cognitive

so when a ministry of foreign affairs or

something is

doing a thing

it is embodied in the sense that it

participates uh in a greater network of

constraints uh which collectively build

organization

uh

which entail the behavior the specific

h-specific decision of the um

ministry of foreign policy so for

example if

there is some kind of

collective

panic moral panic against communism it's

likely in the absence of any

uh formal statement against communism

that

uh decision be taken by that this

shouldn't be best against uh specific

people and words and

themes but urban needs communists

and this is not entailed by anything at

the formal level at the level of
symbolic computation that
could be construed as competing the
highest or most
recognizable forms of state cognition
yet this participates in
a collective behavior
that's adapted
from the border system
this is basically the
nation states distinction familiar
the aggregate behavior of specific sets
of
constitution constraints
past processes will build the strong
processes and
physical entire behavior and this is
um
not something that needs to be
acknowledged from any higher level
and this yes
is uh cognition
and this grounds of view of hemorrhagic
recognition are something that
follows states um as something that is
uh grounded in not the body of any
single individual but in the body of the
state as
the set of its consulting constraints
thanks cervel i also put up a quote that
people can pause the video and read from
the gulag archipelago
that specifically refers to
organizations as organs of the state
which is a little bit of a almost
sinister meat grinder way to describe
the state and it refers to this
interplay between the awareness

and the actions of individuals
and the
systems outcomes at the state level
stephen
yeah i think as well as
this this is really useful i think this
embodiment and our intuitive ability to
embody
um does then recapitulate
the influence process
so as well as these terms
the seats of government or whatever
coming from say we see it coming from
roman literature but over time it's been
encoded and encoded and maybe abstracted
out and we use words and sometimes you
have to look up the etymology to sort of
find out where they came from but we we
do intuitively see a difference if
someone says someone's going to take
their seat at the senate where you get a
feel for that but you don't say they're
going to take their seat at the palace
they're going to take their they're
going to they're going to sit in the
throne right
because the whole point of the palace is
that those seats are not
everyday seats right
so there's there's a but you get that
because of how you know space is treated
around one of those seats
like you don't you don't put a load of
seats around
and roll up a load of thrones you give
space to it um and there's actually a
lot of interesting work in physical
theater where you literally can work

with space
to convey how much space you give
someone
will convey how much power they have
that reminds me of
the term meronymy so that's like when you
say
there were 10 mouths to feed it's
referring to a
system
by referring to the sub part of the
system so it's kind of like marrow
topology is about the parts and holes
and neurology and so the meronymy is when
revealed through our language we say
there were 20 seats on that committee
um
and the fluidity at which that is
semantically unpacked
shows that there is a relationship like
when you say there was 20 mouths you
think there was 20 people
and that includes a lot of our
assumptions um you know 10 pairs of
hands or you know 10 hands
what does that say about our assumptions
about you know ableism or about what the
default
is
in our um way of speaking like that
stephen and then anyone else
yeah uh
and well ableism definitely comes in
then as these these codes as people
start to look at ableism that that's a
very good point
and also there's that intuitive
understanding that you know there's the

idea that a team
generally is really more between five
and ten people to optimally be a team
and then you've got this idea of a group
so you have an intuit when you hit 20
you've gone past a small group into a
large group and there's a whole work
around large group interventions and
getting
aspects of the system in the room which
is about when you can no longer
look at each person in the eye and hold
it which is sort of after 15 16 people
so again you hit this
to some extent
we don't need to be taught that
these things evoke
the reality for us as this structured
organism
in space in place now it's probably
different for a dog on four legs
maybe they can only sniff around four
dogs right and that's a big crowd for
them so you know and is that about their
mind or is that about their
their morphology
so is this definitely an interesting
question
thank you serval and then anyone else
i like the way you talk of the throne as
something that uh whose function is
communicated through medically embodied
um
use
because
the community activity of the state is
embarrassed in
constraints that are not human but

uh those constraints needs to somehow
affect the behavior of humans of because
those fact faster dynamics are basically
the dynamics of a human individual they
need ways to constrain it and influence
it to exist as constraints

and

a major way they do so is by
building the niche humans living
so building the way humans
embody their perceptions
relate to each other to each other take
decisions etc etc

and

to simply build a monumental
architecture to convey that specific
buildings have specific functions and
are very important and that you have to
understand the uh

the underlying philosophy to exist as a
functional member of society it's
a way to do so and that's a way which
the throne

basically affects people to believe in
the kind of exceptional
qualities of the subbrain which are
necessary for a state uh

a state that has a sovereignty to exist

[Music]

so the throne in the crown or or any
other kind of

token there so it's um it's embodied it
refers to a wearable or a bodily
position

it refers to of course in a cultured
and a semiotic
context

it's enacted that relates to declarative

speech acts

i pronounce you married

the speech act is

the

action

and so when we're talking about writing

and language driven systems

which is a key part of again that that

deep histories section

declarative speech acts become very

important when you're served the

subpoena or when something is ruled then

the hammer comes down on the table with

like sort of like the finality of the

act

like an onomatopoeia relative to the

declaration

that was uttered

in a certain ritualized context

um

these are great examples about

how we can use like the 4e framework um

and also i was going to say it's

extended because the crown and the

throne they're literally off of the body

and then what's the story that's like

the total um

uh

like oh king got owned

emperor's new clothes

where the extended cognitive apparatus

that signals the cultural position of

the king

is revealed to be not

what it is

and so it's like

to strip the person of their role it's

just like stripping off their clothes

and these are just all kinds of very
accessible
phenomena
that we can ask
how does this support
a 4e
or an active inference approach
relative to some other approach like
signal processing yeah i'm sure you can
make the neural network that recognizes
a crown versus other images
but
that is like a sub-component of visual
perception
within this broader picture
and then also it helps us ask um what if
and how we could design
embodied and cultured enacted extended
etc
systems
that reflect
the statistical regularities of who we
are now and how we think things are
and how we expect and preferred things
to change
yeah i i think that's a very good point
and
maybe this is where
you know the survivor was talking about
constraints there and also i suppose it
comes back to the idea of when to stop
i'm talking i talked about this a little
bit with dean uh yes
yesterday is um when do we know when to
stop is one of the big challenges about
you know ecological context
and i keep thinking so you know you have
the hammer comes down you know if you

have a state you know the hammer comes
down it
you you you viscerally have an embodied
understanding of what's being
communicated by then
you know that comes down something's
been hammered out and in the houses of
parliament in the uk they have this big
mace they call it the mace it's an
ornament an ornamental mace but of
course amazed well i don't know if this
is exactly true but i i think the mace
normally as something you club someone
at the head with right in a battle right
so
it's there as maybe that that big stick
con idea that's been brought in but why
one thing that really came to mind when
i was when this was coming up was what
happened during nazi germany
when the new technology of sound systems
came out so the big thing that happened
was these big rallies that completely
overwhelm people is no one had had just
like we didn't have with social media
before no one had had an embodied
experience to calibrate
and in some ways you maybe never can
because it becomes slightly beyond the
concepts of embodiment
what it meant to be in this big
arena where you heard sound coming from
everywhere
and it then the idea is that that was
part of the reason why it became
disruptive maybe
we weren't able to constrain and it just
became over a cancerous almost overwhelm

but there's something about new technologies and the ability to constrain and find a state that is to some extent sustainable um and when it becomes out of control and is there something about the embodiment that daniel was talking about there thank you steven serval and then anyone else so i want to emphasize that uh as in their formalism i'm a cleaner too the the constraint is not um does not entail the notion of an interdiction of a tap down order or of um of a stop on a behavior a constraint is a through slow process that modulates that modulates a fast process for example if we agree on the sound of some word say a lamp so we have a relatively slow process that is our couple behavior that will constrain our relative faster processes in the sense that when one of us use the word lamp the other will understand what it refers to so it will basically lower the dimensionality of the possible behavior i have because i will only pick behavior that are consistent with with the fact that you said the word lamp as i understand it so the way that we culturally build our niches

um

it's not

uh

something we understand as constraining

it's something we understand as creative

and yet it corresponds potentially to

the um

the concept of constraint as would be

used by a physicist or stem biologist

which is the sense i use in that

paper

[Music]

thanks

it reminds me of a paper could find it

with like uh it was like developmental

constraints or opportunities

and it was just saying you can think

about like okay i guess there's some

morphological creature that just extends

in a blob

but any other one that has facilitated

enabled affordances to grow in one

direction

can be framed as the constraints from

the other dimension and so there are two

sides of the coin to talk about what is

the constraint that enables it's like

chess

it's the enabling constraints on chess

moves that allow

brilliant chess games and a culture to

actually arise

so survival with the direct response

then stephen and blue

yeah thank you

uh i think blue uh

other hand at first i mentioned uh but

what i wanted to say is that um

to have something that exists as a
meaningful concept to have interesting
behavior

uh you need essential breakings for
example you need

a particular gas to not be independent
the default assumption of a physicist is
that

nothing everything depends on nothing
each particle evolve independently
essentially

so

this means that i need basically
the mean and the variance of their
individual trajectories to

to have all of the insert information on
my system uh this correspond uh to like
pressure and temperature and depending
on

what the trajectory is

um

well this there is no claimation

actually uh whatever

uh

but if i uh somehow uh forbids um

if i somehow build

a collection between the behavior of
those particles then

there are new possible behavior but
emerge you have new possible elements of
this dead space

that exist you have a creation

as far as a physicist that is concerned
with behavior within certain species

um is

as far as uh by phrase but the physicist
sees question because there are things
that were not part of the set base he

drew at the beginning of this system
that become an existence
something that was not in that
now is
and what is interesting is this process
of becoming
us i don't know who uh as we discussed
earlier
uh because uh you have kinds of
symmetries building and symmetry
spracking that are constructive and yet
they need to be constraining because if
everything just arbitrarily adopts any
kind
of behavior in the space that is
accessible that is possible in principle
you have just a philanthropy you have
what noise you have literally nothing
there you have just particles that
blah
and
the facts uh
of existence that we have things to talk
about it entails uh the concerning of
the possible space to
everything is acrylically to some
of the things are more likely and
it entails a meaningful map
between the possible
states at the macro level at the
lowest possible level of description
and the states as
we are interested in in a specific uh at
a specific moment
so thank you i was uh writing on the way
that constraints are not restrictive but
creative
thank you the absence of restraints

statistically constraints would just
refer to you know an unconnected
bayesian graph
just independently vibrating
variables
and so to have anything that's tracking
or anticipating something else
we're discussing the imposition of
constraints using that term stephen and
then andy
you mentioned i said daniel you put that
enabling systems uh slash constraints so
i think
um
there is that i i totally get and i
think it's a very good point about you
know you've got the statistics
stochastic chaos and you've got the
general dynamical processes and there's
there's going to be some way of
constraining that through different
levels
um there's also then
uh you know if we take consciousness as
being at the back the brain stem and
not in the frontal cortex you know
and it's then sending stuff out
you know i can see that biological
dynamic but you also have this enabling
system process of scale i suppose that
in a way i was just referring to and
these things can start to be like the
the ball rolling down the hill so to
speak
you know it starts to accumulate
its own
because it doesn't need to extract
information

to make inferences to make things happen
it's just happening through
thermodynamics and dynamics
so um i wonder how
we can think about the challenge of
these macro systems then
the enabling system overriding the
emerging
dynamical system
thank you
andy

is there a fundamental difference
between thinking like a state and
thinking like a tribe or is it just a
level of sophistication
um with what specialties are are enabled
once you get a state level
thanks for the question sir states
states has administrative systems as a
tribe does not
uh so
like you are
a lot of ways societies can be organized
that are uh somewhat in a continuum
and what i call a state is a specific uh
control system that well is in a
precipitatory state either it exists or
it does not
and in the case of uh
tribes i don't think um
most of what we call tribe of an
integrated decision system more like
shared
ethnic
cultural norms
and if they have intuitive decision
systems it's certainly nothing like uh
administration

it's so fascinating because the tribe
can be implicitly existing without a
state as cerval just noted
and it makes me think about
web 3 and taos where it's almost like
the states are being defined first and
then the people are rushing in to fill
so instead of the historical emergence
of specialization and elaboration
and
formalization on top of an implicit
background of the people
and then in a complex interplay as we've
been discussing this is almost like a
computational state that people are
migrating to
with a preset governance system
sometimes one that's incredibly
constraining and people may not even
know how constraining
until they're there
so that's an inversion
of
how we've seen
the um members of the state and the
state's evolution engage so more things
change the more they stay the same but
it's a different river for sure so dean
blue stephen
i think i found something that i can
i can say i can agree with
with
serval and and maybe some of the people
who really like genetics because i
i think constraint really matters and i
i think the idea that ratio
is not phenotypical drift through
uh neutral mutations even though

mutations are a ratio of the

non-mutations

i also think that ratios and randomness

accommodations

are not the same thing so i think

constraint

allows us to be able to see the

difference between those two things

one is when a triangle is a triangle and

the other is when a variation plays upon

a variation and i think we can accept

both

that's what i think the constraint piece

actually helps us

pull the two things together

i thank you dean blue then stephen

so just to kind of go back to what um

andy was saying about the difference

between a tribe and a state but also

like

when do we know when to stop um and

processes that that constrain and when i

think about this like bi-directional

information flow

like if the emergent energy is too

strong

like it then it's out of control right

there's no constraint but but like when

the constraint is too strong like i

think about like you know even in mental

action the idea that like you have your

grip is too tight and so like what

whatever you're trying to control like

pops out of your grip right like so like

you can squeeze a balloon and then boom

um and so there has to be like some

balance and i just wonder if that

balance is like quantifiable like does

downward information have to equal
upward information flow like
is there like a balance between those
two things that allows like a
self-emergent or not a self-organizing
entity to maintain in the state of an
attractor because if like the
information flow is too much in one way
or the other
the balance goes off and it it'll steer
away from that attractor space
steven and then i'll add a comment on
that blue
yeah i think that there's the the
ability to constrain
is going to be at different levels in in
the state
you've got certain things which are just
like
constitutional you hear this in the
states there's constitutional rights
right they're fixed they can't change
it's been there for 200 years or
whatever and then you've got more
negotiation and in the uk you've got the
government has to go and talk to the
queen every couple of weeks or every
week so in theory the queen could
constrain now again they don't just have
to send a letter they have to go and
meet in person and talk there has to be
so then there's this inter-subjective
dynamical constraint and then of course
then for scaling you've got the kind of
we can't really do it as organisms so we
put in an electoral system but the
electoral system
and the the democracy in north america

comes a lot from the tribe that the the
sort of scaled tribes of north america
as much as it does from france it's
quite interesting and one thing they
have in toronto we have the treaty of
the dish of one spoon and the treated
disher one spoon is basically they sat
round

in a circle and they have a spoon they
have a dish with one spoon and you only
take enough for yourself and you leave
enough for nature and it's past rounds
and that's the treaty of how to look
after the great lakes

and actually it was a lot of that kind
of

slightly scaled version of tribal work
which then informed and said okay
europe you're killing each other all the
time what about something a little bit
like this

so actually a democracy in the u.s was
partly from the euro

not really tribes anymore but that kind
of scaled peace so i think um

there's there's

there's that need to sort of draw back
into something which is more contextual
more dynamical um as well as the kind of
i don't know the more kind of

law based approach

thank you so people have been mentioning
the getting a grip which was of course
from the

paper

about how states get a grip on their
subjects and that was drawn from the
work of of yella and ritfield

talking about optimal grip and
skilled performance and all these other
cool
inactive and ecological psychological
ideas we've talked to so it's like of
all the trajectories that the golf club
can take
we're constraining that to some lower
dimensional manifold that's getting a
grip
and i totally agree i think a good
example of an empirical system
and about
what kind of reciprocal interactions
exist and what attractor states does
that beget
uh is the federalist anti-federalist
discussion at the end of the 1700s in
the united states
with people saying if you have the
affordance for a standing army at the
federal scale
then we'll see this happen at the lower
state level or if there's the ability to
engage in foreign diplomacy or print
debt at the federal level
then we'll see this happen or not at the
state level and so i think that's where
we go from sort of the
what has been and what is to the future
that we don't know about the what-if
where
we are in that scale-friendly
level where we're talking about
designing
or modifying or perturbing or
intervening in
those causal threads that do link

and co-constrain
the smaller sub-units with the larger
processes
and
how do we
talk about the space of design there
and integrate our preferences and
expectations
together with just
sometimes the um
dazzling and tragic
and complex history of just
everything we see on this map
so we'll have just um a few minutes left
let's from anyone who would like to
let's just hear like some last thoughts
what's something cool from today or
what's something that you're looking
forward to discussing more next week and
otherwise it's been an awesome
dot one so stephen and then we'll go
around to anyone else
yeah following on what you were saying
there daniel and i'll use this as a sum
up it slightly opens up a new thing
which maybe we'll get to the next
session but
you know how do we see what happens with
the nature of tech companies and the
huge affordances that that puts into our
landscape
also the constraints if we take the
mcluhan tetrad of how it then changes
the way we behave overall
they're introducing
rather than
um a constraining
or a

sort of a wise constraining way
it's it's putting out there what's out
there and it's it's affording huge
amounts of new
novel interactions be it microsoft be it
google be it
facebook
and then they're getting hugely
powerful just through
permeation into society so uh you know
something that's interesting is then
where does
you know i think that may reflect one of
the challenges we have at the moment
with corporations getting so big
so thank you and this is a very very
good talk and
lots of things are sparking off it so i
appreciate it
thank you yeah like
nation states used to be like states
were the big players in town
there was big ones and there were
smaller ones but that was sort of like
the ultimate level of analysis
web 2
we saw
websites stock tickers apple amazon etc
become bigger than many states and
intertwined and amongst them in a way
that sometimes is even more advantageous
though they don't have citizenships
themselves they have user identifiers
they have revenue flows they have
governance models
and then now
into web3 we go
and platforms and protocols maybe even

decentralized ways
of accumulating governance and economic
systems
that will also
be
at least initially
larger than smaller nation states and
web to tech companies so there's gonna
be like
legacy like web zero one two three four
all of these
systems are going to be
on the same territory
but they're going to have different maps
so they're going to be coordinating
very differently
um would anyone else like to give a
closing
thought i think there's a lot of stuff
in the dot too that we'll be able to
like jump off into like survival next
week i want to
hear about where we go next like what
are your next personal research
goals and what are other
opportunities for individuals who want
to like learn more or research or apply
in this area or under this framework
blue and then anyone else
so just a final thought i'd like to like
discuss more or maybe get more into like
what is the state anticipating i mean
how do they anticipate uh what's coming
next just predictive behavior
yeah and it will be fun also to like
just connect some of these to
the core terms in the active in
anthology just like what are the sense

states of the states what are the action

states of the states

let's you know what is attention is that

the most wanted list

what are the ways that we can just take

the terms that we already have and start

to build up this table of mappings

so that we can and maybe it's gonna fit

on multiple levels so it's not gonna be

just the one and only

map of maps but it'll be a draft

versionable participatory set of

mappings and that'll be fun

dean

yeah what i really appreciated about the

paper was the ability to sort of look at

things through the scale in variance

lens and now i'm really curious to see

what would happen if we got off the map

it started moving around and then

actually incorporated scale variants

because i think that's really the and

then what happens piece that this paper

of paper forwards it gives us a good

grounding and now we can step off of

that and see

to think in the hypothetical space i'm

really curious to see where that goes

excellent

and if you have any last comments

well thanks for letting me play i really

enjoyed the conversation

um and a lot of this was new to me right

i'm new to the even the idea of active

inference but being able to discuss it

in concepts of the state uh

incredibly interesting so thanks for

letting me participate

excellent well it's always a two or more
way street so we all had an awesome time
thank you everybody for
participating we hope to see you here
and or participants through listening
in another live stream in the lab
working with us on projects so
see you all later talk to you next time